Project Title: InsureGuard — AI Insurance Fraud Detection System **Project Design Phase-I** **Team ID: 699d5f83fdf409fdbd1bec82**

Define CS, fit into CC

1. CUSTOMER SEGMENT(S) CS

Who is your customer?
i.e. working parents of 0-5 y.o. kids

- Auto Insurance Companies
- Claims Adjusters and Risk Analysts
- Dedicated Fraud Investigators

6. CUSTOMER CONSTRAINTS CC

What constraints prevent your customers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices.

- Regulatory/Time Constraints: Strict legal deadlines require fast claim payouts, preventing deep manual investigations for the vast majority of claims.
- Financial Constraints: Insufficient departmental budgets to hire armies of dedicated human fraud investigators.
- Technical Constraints: Standard claims adjusters lack machine learning expertise, requiring any AI solution to be hidden behind a highly intuitive visual interface.

5. AVAILABLE SOLUTIONS AS

Which solutions are available to the customers when they face the problem
or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. pen and paper is an alternative to digital notetaking

- Manual Auditing: Leverages human intuition but is extremely slow, expensive, and severely prone to fatigue and error at scale.
- Rule-Based Software: Fast but highly rigid. It generates too many false positives and is easily bypassed by sophisticated fraudsters who know the rules.
- Private Investigators: Highly thorough but prohibitively expensive and slow, making it viable only for a tiny fraction of claims.

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

2. JOBS-TO-BE-DONE / PROBLEMS J&P

Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides.

- Accurately filter and identify fraudulent auto insurance claims from a massive daily volume.
- Reduce massive financial losses caused by undetected fraud payouts.
- Accelerate the approval process and payout times for legitimate, honest policyholders.

9. PROBLEM ROOT CAUSE RC

What is the real reason that this problem exists?
What is the back story behind the need to do this job?
i.e. customers have to do it because of the change in regulations.

- Fraudsters continually evolve, using highly sophisticated tactics (like staged accidents) that look completely legitimate to human reviewers on paper.
- Human reviewers are simply unable to simultaneously process and correlate 30+ complex variables (like age, incident time, and claim-to-premium ratios) to spot hidden mathematical anomalies across thousands of claims.

7. BEHAVIOUR BE

What does your customer do to address the problem and get the job done?
i.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace)

- Claims adjusters spend hours manually cross-referencing incident data (like time and collision type) against property damage and police reports.
- Investigators dig into a claimant's history to find suspicious patterns or previous claims.
- For highly suspicious cases, companies hire expensive third-party private investigators to physically verify details.

Focus on J&P, tap into BE, understand RC

Identify strong TR & EM

3. TRIGGERS

What triggers customers to act? i.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news.

- A sudden spike in claim submissions (e.g., after a severe weather event).
- Quarterly financial reports showing rising losses due to fraudulent payouts.

4. EMOTIONS: BEFORE / AFTER

Howdo customers feelwhen theyface a problem or a job and afterwards?
i.e. lost, insecure > confident, in control - use it in your communication strategy & design.

- Before: Overwhelmed by massive claim volumes, frustrated by continuous financial leakage, and anxious about missing hidden fraud patterns during rushed manual reviews.
- After: Confident in AI-backed approval decisions, relieved by the automated reduction in manual workload, and in control of the investigation pipeline.

10. YOUR SOLUTION S

If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality.
If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.

- InsureGuard is an automated machine learning web application that uses a tuned Random Forest classifier to instantly evaluate claims.
- It analyzes 36 distinct features and translates complex AI predictions into an intuitive Streamlit dashboard with an animated 0-100% risk gauge.
- It acts as an intelligent, lightning-fast first line of defense, matching the technical limitations of adjusters by providing clear visual insights rather than raw code or data.

8.CHANNELS of BEHAVIOUR CH

8.1 ONLINE
What kind of actions do customers take online? Extract online channels from #7

8.2 OFFLINE
What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.

8.1 ONLINE: Adjusters use internal claim management systems, search public DMV databases, and run digital background checks on claimants.
8.2 OFFLINE: Investigators make phone calls to local police to verify reports, interview witnesses, and conduct in-person inspections of vehicle damage.

Identify strong TR & EM